

End-to-End Telecom Customer Churn Analysis & Prediction

1. Executive Summary

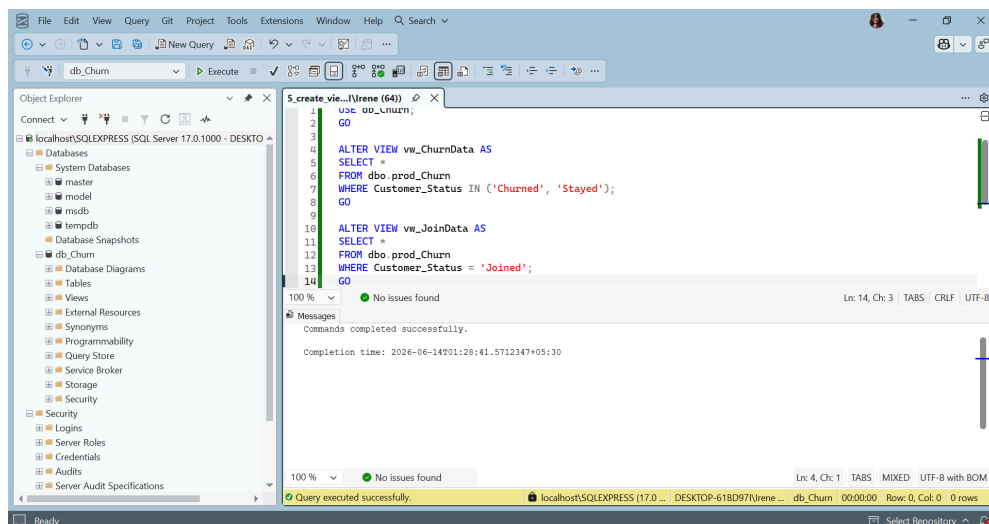
In the highly competitive telecommunications sector, the cost of acquiring a new customer significantly outweighs the cost of retaining an existing one. This project encompasses a complete, end-to-end data pipeline designed to analyze historical attrition patterns, engineer a machine learning prediction model, and deploy interactive business intelligence dashboards.

By applying a Random Forest classification model to live customer data, this pipeline successfully identified **1,270 active accounts at high risk of churning**. At an estimated average monthly charge of \$65, this represents over **\$82,000 in monthly recurring revenue** isolated for targeted retention campaigns.

2. Data Architecture & ETL Pipeline (SQL Server)

A robust backend was established using Microsoft SQL Server to ensure data integrity and automate the transformation process from raw flat files.

- **Data Ingestion:** Raw historical data was imported into a staging environment (**stg_Churn**).
- **Data Cleansing:** Executed exploratory queries to identify null distributions and standardize categorical features (e.g., standardizing **Bit** datatypes to **Varchar** to prevent load errors).
- **Production Schema:** Engineered a cleaned production table (**prod_Churn**) handling null values contextually (e.g., converting null 'Premium Support' to 'No').
- **Data Isolation:** Created distinct SQL Views (**vw_ChurnData** for historical training data, **vw_JoinData** for active customers) to prevent data leakage during model training.

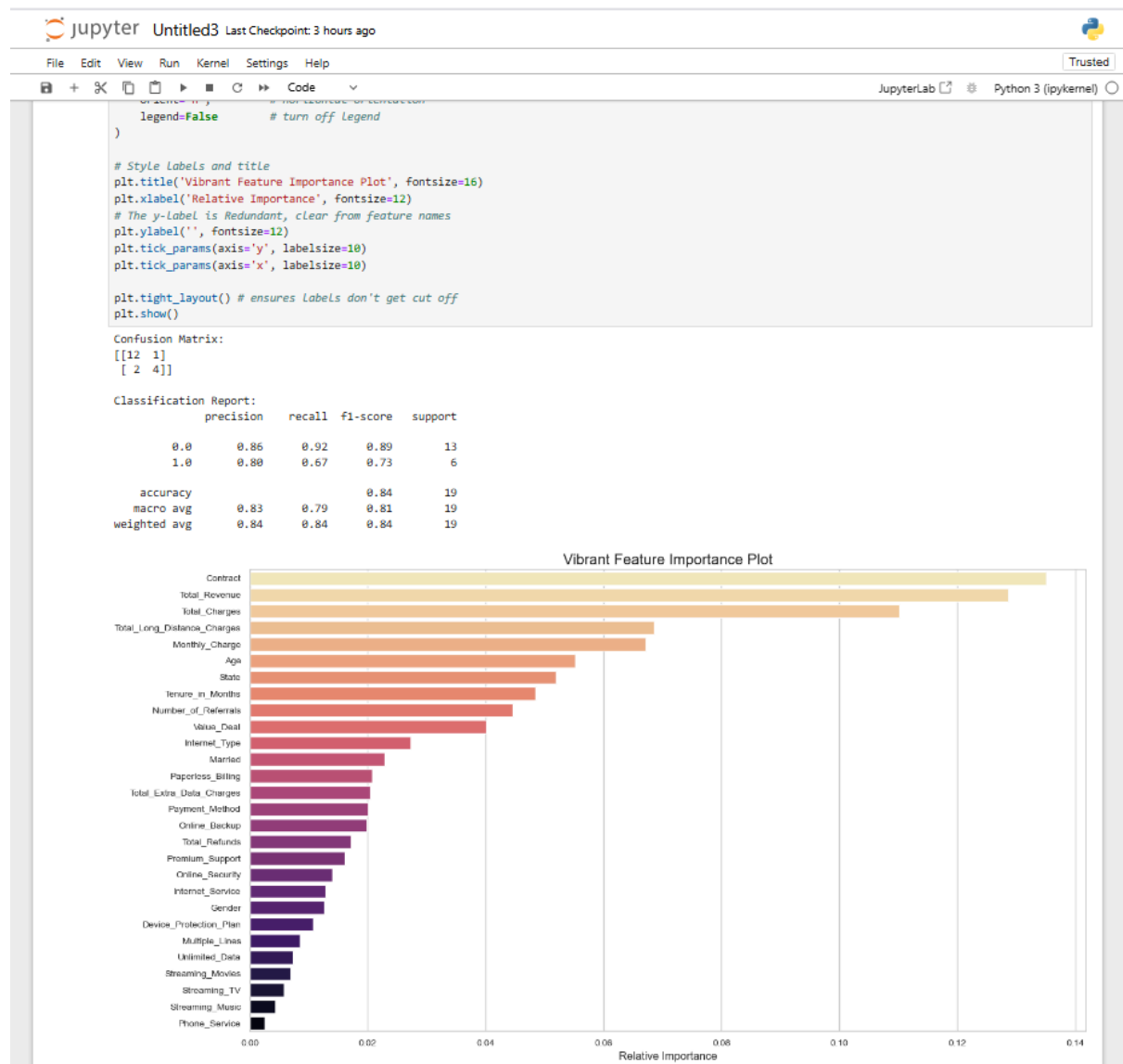


3. Predictive Modeling (Python & Scikit-Learn)

To predict future churners, a **Random Forest Classifier** was trained using Python in a Jupyter Notebook environment. This algorithm was selected for its robustness against overfitting and its ability to handle complex, non-linear relationships across 19 categorical demographic and account features.

Model Engineering & Edge-Case Handling:

- **Preprocessing:** Categorical variables were encoded using [LabelEncoder](#).
- **Defensive Programming:** Real-world data pipelines frequently encounter data drift. During the scoring of new data ([Customer_Data.csv](#)), the model encountered previously unseen categorical labels (e.g., a new geographic state). I engineered a dynamic safety net within the pipeline to temporarily map unknown labels to the known class baseline, preventing pipeline failure and ensuring continuous prediction delivery.
- **Output:** The model scored the active user base and exported the at-risk cohort into [Churn_Predictions_Report.csv](#).



4. Business Intelligence & Reporting (Power BI)

The structured SQL views and predictive outputs were integrated into Power BI to create dynamic, stakeholder-facing dashboards.

- **Calculations (DAX):** Authored custom measures to calculate dynamic Churn Rates, Total Active Customers, and Predicted Churn Counts.
- **Mapping:** Implemented custom mapping tables ([mapping_AgeGrp](#), [mapping_TenureGrp](#)) utilizing one-to-many relationships in the Power BI Model View to enable complex cohort filtering.

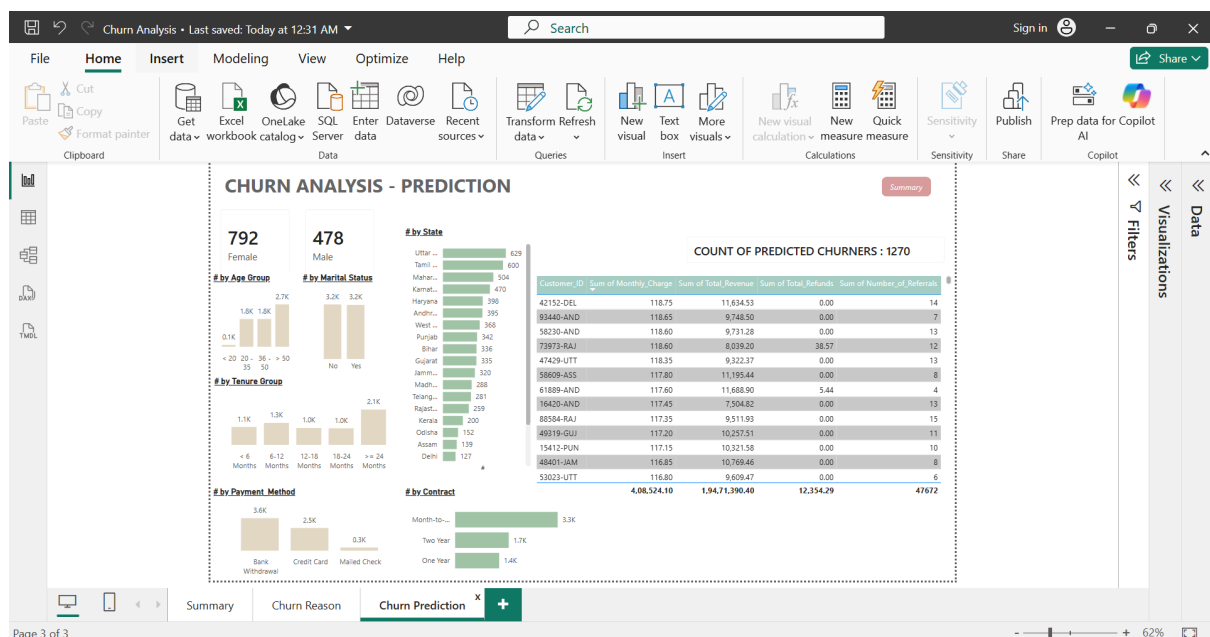
Dashboard: Historical Summary

This view allows stakeholders to dissect the root causes of past customer attrition. **Key Finding:** Customers on month-to-month contracts utilizing manual payment methods (mailed checks) exhibit the highest historical churn variance.



Dashboard: Predictive Action Center

This view operationalizes the Machine Learning output, providing the sales and retention teams with actionable targets. **Key Finding:** Lack of supplementary services, specifically 'Online Security' and 'Premium Support', is a dominant predictor of future churn among the 1,270 flagged accounts.



5. Strategic Business Recommendations

Based on the synthesized data and predictive modeling, I recommend the following immediate actions to reduce attrition:

1. **Targeted Contract Upgrades:** Deploy an automated email campaign to the 1,270 predicted churners offering a 10–15% monthly discount incentive to transition from a Month-to-Month contract to a 1-Year commitment.
2. **Service Penetration Strategy:** Offer a 3-month free trial of 'Online Security' and 'Device Protection' to at-risk accounts. The model's feature importance indicates that integrating customers into the wider product ecosystem significantly increases retention probability.
3. **Payment Automation:** Introduce a one-time bill credit for high-risk customers who switch from manual check payments to automatic Bank Withdrawals.